Semi-Supervised Labels and Outcome Prediction for Clinical Trials

CS5785 / ORIE5750 / ECE5414 — Midterm Project

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Abstract

Part 1 asks why a trial stopped early. Only 157 of 1,171 training trials have visible reason labels. We use the 150 supplied reason embeddings. An anchored classifier combines logistic regression, neighbors, and shrinkage discriminant analysis with clustering-assisted self-training. A second pipeline adds graph, cluster, and neighbor information. A fixed 80/20 blend scored a reported Kaggle sample-wise F1 of 0.89935, compared with 0.89636 for anchored alone.

Part 2 asks whether a trial meets its primary endpoint. The score is log loss, and the test trials start later than the training trials. We upweight recent trials, average in a neighbor model that also prefers recent matches, and mix in 30% of an earlier text-and-tree ensemble that errs on different trials. The reported Kaggle log loss is 0.58465. A blend that looked better on our historical checks scored 0.58492, so we kept the first result.

Team responsibilities. All three authors worked on both parts and on this report. **Michael Mazzola** led the Part 1 model. **Tarun Mandalapu** built the first Part 2 classifier, and **Anirudha Hadagali** led the later Part 2 tuning. Tarun Mandalapu drafted the longer project notes that this summary condenses.

1. Part 1: stopping reasons

1.1. Data and the score

The files hold 1,561 trials that stopped early [2, 9]. Training has 1,171 trials: 157 with real labels and 1,014 with every reason marked -100, which means unknown. Validation has 78 labeled trials. Test has 312 later trials, from 2010 to 2016. A trial can carry more than one of Safety, Funding, Efficacy, Recruitment, and Other. Among the 235 real labels, Recruitment appears on 64% and Funding on 5%. The labeled trials are mostly early, so those rates may differ in the test set.

The selection score is sample-wise F1, the average of per-trial F1. We also report each reason's F1 and their macro average. One validation trial can move sample-wise F1 by about 1.3 percentage points. Later comparisons therefore use eight repeats of five grouped folds on the 157 training labels and on all 235, plus three earlier-to-later windows totaling 98 predictions. Embedding-duplicate groups stay together, and masked duplicates of held-out groups are purged. These splits were reused during development.

1.2. What the model does

Each trial has 150 supplied stopping-reason embedding coordinates. IDs align rows and are not features; years define chronology checks. On the earlier self-training model, extra start year, eligibility text, disease, or drug features [4, 5, 7] did not improve validation sample-wise F1: embeddings alone scored .893, versus .891 with year, .876 with eligibility or disease text, and .889 with fingerprints. These are historical feature checks, not ablations of the final blend.

The *anchored* component standardizes embeddings using permitted gold and masked training rows. A 30-component diagonal Gaussian mixture uses 30 principal components [1, 6], with k-means initialization, three restarts, and seed zero. Component reason profiles average the gold labels with one pseudo-count of the overall rates. The pseudo-label teacher mixes 5% of these profiles with 95% of the classifier score. Earlier k-means and DBSCAN checks were weaker on Kaggle. DBSCAN-plus-logistic reached .910 validation F1 under an earlier protocol, but its original and 235-label submissions scored .87500 and .86452 respectively.

The anchored classifier combines multiclass logistic regression ($C = .1$), cosine 10-nearest-neighbor votes, and shrinkage linear discriminant analysis (LDA), with fixed weights .5/.3/.2. All use the 150 standardized coordinates. A multi-label trial is duplicated across its positive classes for the multiclass fits. Logistic weights divide its mass across positives and use class weights fixed from gold labels. Neighbors vote for every positive label, weighted by nonnegative cosine similarity squared. LDA uses equal class priors.

Two pseudo-label refreshes give three classifier fits in total [8, 11]. A masked trial is accepted when its best teacher score is at least .8. Each refresh rebuilds from gold plus the currently accepted rows. Generated rows receive logistic weight .25; the neighbor and LDA components use them without that downweighting. The final 235-label anchored fit accepts 757 masked rows in the last refresh. Gold labels remain fixed.

The second component, *Cand2*, models eight fixed label combinations. It adds soft eight-cluster k-means membership and 20-neighbor label-vote features to standardized embeddings. Its Correct-and-Smooth labeler [3] uses a symmetric 10-neighbor graph on scaled, L2-normalized 64-dimensional PCA coordinates. Correction and smoothing coefficients are .8 and .419962. Generated labels require confidence .5 and receive weight .0630725 in balanced, weighted shrinkage LDA. The strict adapter refits preprocessing, graph labeling, and calibration within inner grouped partitions, with duplicate purges. Temperature calibration uses inner held-out predictions. We average five seeds and convert combination probabilities into five reason marginals.

The selected rule combines anchored raw scores and Cand2 marginals:

$$s = .8s_{\text{anchored}} + .2p_{\text{Cand2}}.$$

We output every reason with $s_j \geq .3$ and always include the highest-scoring reason. These scores are not established calibrated probabilities. The blend is fixed; it has no learned final combiner. Standalone Cand2 used expected-F1 decoding, which is not the blend's decision rule.

Both components use only permitted training inputs under the strict adapter; test embeddings enter prediction only. Final export refit anchored on all 235 gold labels plus the 1,014 masked rows and reused the verified five-seed Cand2 final scores. The .891 validation score belongs to a separate 157-label fit, not independent validation of this final artifact. The anchored notebook is a component account; the frozen blend exporter specifies the complete winner.

2. Part 2: trial outcomes

2.1. Data and the score

Each row is one trial in one phase. A combined-phase trial appears twice and those rows stay together in every split. Training has 8,037 rows through 2013, validation has 1,013 rows from the same years, and the test file has 1,500 later rows, 500 per phase, from 2013 to 2022. Kaggle averages the three phase log losses. We select on that same average. F1 at 0.5, ROC-AUC, and precision-recall AUC are diagnostics. Part 1 reasons are joined only after prediction, for the error table. They are not model inputs.

2.2. What the model does

The baseline is logistic regression on eligibility text: word and bigram counts, at most 35,000 terms, minimum document frequency 3, sublinear counts, three phase indicators, and $C = 2$. Separate phase-only fits overfit Phase I. Drug fingerprints did not reliably help, and a matched drug is less common in test (39%) than in training (46%). Naive Bayes ranked trials but was far too confident for log loss.

The model we built on is a stack of eight classifiers [10]: logistic models on eligibility text, diseases, drug names, codes, drug annotations, and metadata, plus naive Bayes and a 50-neighbor model on the eligibility text. A second logistic regression learns the weights from out-of-fold predictions, five folds, with a trial never split across folds. Recent trials get more weight, with a three-year exponential time constant, so a trial loses about half its weight every two years. That stack, refit on all 9,050 labeled rows, scored 0.58688. The same stack without recency weights scored 0.58810. Our first ensembles had used only the 8,037 training rows, so part of the earlier gap was the extra data.

The neighbor piece inside the stack ignored time and pulled estimates toward a 50% success rate. File 10 does not replace the stack. It averages it, half and half, with a second stack that decays faster (two-year time constant) and that also sees eight neighborhood estimates: several sizes, sharper similarity, and similarity that includes disease, drug, and ICD text. The weights are fixed. This improved every time-ordered window and scored 0.58609, even though the random validation log loss got slightly worse.

File 10 is weaker when the years are mixed. Our earlier ensemble is weaker on later years. That ensemble is half three logistic models, a quarter a depth-6 tree, and a quarter a depth-8 tree. We first added 20% of it (Kaggle 0.58474). Raising that share to 30% gained only another 0.00009, so we stopped. The submitted probability is

$$0.70\,P_{10} + 0.30\,P_{\text{early}}, \quad P_{10} = 0.50\,P_{\text{recency}} + 0.50\,P_{\text{fast neighbors}}.$$

Supervised pieces are refit on all 9,050 labeled rows after the recipe is frozen. Test labels are never used. In the inherited stack, word lists may see unlabeled test text. That changes the features, not the answers. A later blend with an extra tree and a phase trend looked better locally and scored 0.58492, so we did not replace file 13. One upload slot was left unused.

3. Results

Table 1 compares Part 1 recipes on the same grouped splits. The selected anchored/Cand2 blend has the highest reported Kaggle score; the prior anchored/LDA blend retains the higher Pooled235 mean. Against anchored, the winner's repeat wins/ties/losses were 5/1/2 on Train157 and 4/1/3 on Pooled235. It missed our original five-wins-in-both-pools

Model	Train	Pooled	Val.	Time	Kaggle
80/20 Cand2	**.9352**	.9216	.891	**.888**	**.89935**
75/25 LDA	.9321	**.9234**	.891	.884	.89850
Anchored	.9321	.9214	.891	.884	.89636
LDA ensemble	.9318	.9216	**.906**	.878	.89401
20% PySR	.9275	.9198	.885	.878	.89423
20% kernel	.9334	.9209	.887	.871	.89316

Table 1. Part 1 sample-wise F1, higher is better. Train and pooled average eight grouped repeats on 157 and 235 labels; validation uses 78 trials; time pools three windows. Cand2, LDA, and kernel rows blend with anchored. PySR's parent is the earlier 75/25 anchored/LDA blend.

Model	Time	Grouped	Kaggle
Phase-specific ensemble	—	—	0.59562
Text-heavy ensemble	—	—	0.58993
Stack, no recency	0.5421	0.5797	0.58810
Recency stack	0.5374	0.5874	0.58688
File 10, recent neighbors	0.5359	0.5869	0.58609
File 13, 30% early ensemble	0.5334	0.5790	**0.58465**
File 15, tree and trend	**0.5327**	**0.5783**	0.58492

Table 2. Part 2 log loss. Time averages the forward windows. Grouped is trial-grouped cross-validation. The first two submissions used an earlier local protocol, so those cells are blank.

advancement rule. An exploratory manual trial then supplied the best reported Kaggle result; we preserve that selection history. Local blend scores recombine retained held-out component predictions, rather than refitting both components for each blend.

The winner's 78-row validation labels match anchored. Per-reason F1 is .941 Safety (9 positives), .667 Funding (4), .759 Efficacy (14), .990 Recruitment (50), and .471 Other (10): sample-wise .891 and macro .765. These are development results, not hidden-test label scores. Kaggle values are user-reported; the leaderboard partition and actual uploaded bytes were not independently checked. Overlapping repeats do not establish independent statistical significance.

Table 2 is the Part 2 path. Lower log loss is better. File 13 is the submission. File 15 wins both local columns and loses on Kaggle, which is why the local ranking was not enough.

Table 3 is the supplied validation split, with preprocessing fit only on that split's training rows. The recency stack improves Phase III log loss from 0.529 to 0.516 and Phase I F1 from 0.765 to 0.793. It does not improve Phase II F1. File 10 is slightly worse than the recency stack on this random split and better on the later-year windows in Table 2. That is why a higher validation F1 was not the submission rule.

Table 4 counts failed validation trials that the outcome model called successful. The ensemble is the early text-and-tree model that later became 30% of file 13. The readable groups are Phase II Recruitment and the three no-reason groups. Phase III failures are called successful much more often than Phase II failures. Most reason tags are Part 1 predictions. The later neighbor model still called all 9 Phase III efficacy-tagged failures successful.

Phase	Metric	Baseline	Recency	File 10
I	F1	0.765	**0.793**	0.762
	ROC-AUC	0.783	**0.794**	0.787
	PR-AUC	**0.835**	0.830	0.824
	log loss	0.556	0.556	0.560
II	F1	**0.657**	0.647	0.638
	ROC-AUC	0.740	0.753	**0.754**
	PR-AUC	0.717	**0.732**	0.730
	log loss	0.600	0.593	0.593
III	F1	**0.833**	0.825	0.829
	ROC-AUC	0.774	**0.790**	0.786
	PR-AUC	0.862	**0.882**	0.877
	log loss	0.529	**0.516**	0.521

Table 3. Part 2 validation by phase. F1 uses a cutoff of 0.5. The other rows use the probability of success. Phase sizes are 149, 501, and 363. Best value per row in bold. Tied log losses are left unbolded.

Phase	Reason	n	Base	Ens.
I	Safety	9	.56	.44
I	Funding	4	.50	.50
I	Efficacy	2	.50	.50
I	Recruitment	22	.36	.27
I	Other	5	.80	.80
I	No reason	28	.43	.39
II	Safety	15	.40	.40
II	Funding	4	.00	.25
II	Efficacy	10	.30	.30
II	Recruitment	87	.20	.16
II	Other	18	.33	.28
II	No reason	144	.35	.34
III	Safety	6	.50	.67
III	Funding	0	—	—
III	Efficacy	9	.89	.67
III	Recruitment	21	.52	.57
III	Other	8	.63	.75
III	No reason	82	.56	.55

Table 4. Share of failed validation trials predicted successful. "No reason" means the trial is absent from Part 1. A trial with several reasons is listed under each.

4. Why these choices

For Part 1, clustering supports missing-label generation, while supervised scores provide most of the prediction.

Standalone Cand2 scored .89102, below anchored's .89636, but its small contribution raised the blend to .89935. This supports complementary errors for this fixed combination, not standalone superiority or a guarantee that greater diversity improves performance.

Later trials did not improve the winner. Log-odds pooling scored .88888 despite small local gains; changing treatment of secondary Other labels scored .86645; a diversity-penalized kernel blend scored .89316; and bounded symbolic regression scored .89423. Learned combiners also failed local comparisons. The proposed derived ranking metric J disagreed with some observed Kaggle orderings. Its small selected historical panel and different split boundaries do not establish a general ranking guarantee.

Funding and Other remain weakly measured: validation contains only four and ten positive trials respectively. Extra-feature failures and chronology results are evidence within this development process, not proof that features cannot help or that the temporal checks represent the hidden test population. The selected result improves reported Kaggle F1 by .00299 over anchored and .00085 over the prior blend, with no claim of statistical significance.

In Part 2, log loss and F1 disagree on purpose. A model that moves an uncertain trial across 0.5 can lose F1 and still assign the better probability. The test file is a later, more successful cohort than the random validation split, so we trusted time-ordered checks. Recency weights and recent neighbors are two ways of saying that an old similar trial is a weaker example. The 30% older ensemble is there because it misses different trials, and the second 10% of that share was only worth 0.00009 on Kaggle. After file 15 lost on the leaderboard, a better historical average was no longer a reason to upload.

The reason table says where outcome errors live, not how to tune a cutoff. Failed Phase III trials often ran to the end and simply missed the endpoint, so the eligibility text looks like a success. Recruitment and funding failures are more administrative, and the ensemble catches a few more of them. Safety, Funding, and Other are small in every phase. Only Phase II Recruitment and the no-reason groups are large, and the no-reason group is a lookup rather than a prediction.

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